

Speaker Notes

Monte Carlo Counterparty Credit Risk Engine — Final Client Readout

63 slides

1 INTRO

Monte Carlo Counterparty Credit Risk Engine

Open with who we are and what this readout covers -- a validated Monte Carlo CCR engine built end to end for the ESF derivatives book.

- Berkeley MFE industry project for Capitolis Risk, Quant & Technology -- final client readout, not a status update
- Valuation date 2026-08-28; everything in this deck is reproducible from that single date's market data
- Housekeeping: ~45-50 min including Q&A; happy to take questions as we go on anything that needs more depth

TIME: ~30 sec

2 INTRO

Portfolio MPE99 of \$51.0M on the Brief's Close-Out Definition

We built, validated and stress-tested the full CCR engine the brief asked for; headline portfolio MPE99 is \$51.0M, concentrated in the first four months.

- Don't read the four KPI tiles aloud -- they're there for the audience to see. Instead say what they mean together: the book is short-dated and lightly diversified, so the headline number is really a statement about one counterparty's one trade, not a smoothed portfolio average
- The gap between \$129.8M (today's full uncollateralized exposure) and \$51.0M (the brief's close-out MPE99) is the single most important number relationship in this deck -- it's entirely the effect of daily margining, and it's worth pausing on because everything downstream (CVA, capital, stress) depends on which of these two conventions applies
- This is the 30-second version of the whole 50-minute deck -- if someone has to leave after this slide, they should walk away knowing the number, why it's concentrated, and that margin terms are the open question

TIME: ~1.5 min

3 INTRO

The Book: 16 Trades, Dominated in Size by One \$500M Bond Forward

Show the actual book before anything else -- every number in this deck traces back to these 16 rows.

- Grounding move: before any model talk, make the audience see that this is a real, small, lopsided book -- not an abstract portfolio. That context changes how they should read every later result (a 'portfolio' number here is mostly one trade's behavior)
- If asked why NPVs differ so much in sign and size across similar-looking Equity TRS trades: it's spot moves since inception, not a modelling artifact -- each one was struck at a different time against a different basket
- Useful transition line into the next slide: 'given this is the book, here's the shape of the whole engine we built to risk-manage it'

TIME: ~1.5 min

4 INTRO

16 Trades, 3 Counterparties, 37 Equities and USDJPY, One Engine

Set scope and agenda; this is a readout, not a first look -- we move quickly through concepts the audience already knows.

- Say explicitly that we'll move fast through anything covered in the kickoff deck or prior meetings -- this audience has seen the brief and the basic asset classes before, so dwelling here would waste the time budget
- The agenda order is deliberate: data and pipeline first (so results are trusted before they're shown), modelling choices and trade-offs before feedback (so the feedback section reads as closing loops, not introducing new material), results last and longest
- If the room seems impatient to get to numbers, it's fine to say 'results start in about 15 minutes' and keep moving -- the earlier sections exist to make the results credible, not to be lingered on

TIME: ~1 min

5 BRIEF

What Capitolis Gave Us, and Asked For

Section divider -- the brief and our delivered scope.

- Keep this short -- its only job is to signal we're about to prove delivery against the brief's own words, not our own framing of success

TIME: ~20 sec

6 BRIEF

The Brief: No Initial Margin, 10-Day Close-Out, Capitolis Is Seller

Restate the brief's own assumptions exactly as given, because every modelling choice downstream traces back to them.

- Make the point that we didn't choose the close-out definition or the no-initial-margin assumption -- these came from Capitolis, and we're showing them here specifically so nobody mistakes a given constraint for a modelling choice we made
- The 'strictly the seller' point is worth explaining out loud: it means the book's risk is one-directional by construction (we never benefit from market moves the way the counterparty does), which is why every result in this deck is framed as counterparty exposure to us, never the reverse
- Flag that scenario count being 'our choice, with evidence of convergence' is exactly the freedom we used to justify the N=1k/5k/10k tiering shown later in Trade-offs -- it's not an oversight that we didn't pick one fixed number

TIME: ~1.5 min

7 BRIEF

Every Ask Mapped to What We Delivered, Including Both Extra-Credit Items

Walk the brief's own checklist against our delivery, ending on the two extra-credit items both being complete.

- Move quickly down the rows -- the point of this slide isn't to re-read each cell, it's to let the room see for themselves that every row has an entry, which is the credibility payoff
- Spend the real time on the last row (risky bonds + CDS): be upfront that we could not source CDS data despite trying, and that the risky-bond proxy is a deliberate, disclosed substitution, not a silent gap -- this honesty here builds trust for the limitations slide at the end
- If asked why SA-CCR appears when it wasn't named in the brief: it's the natural regulatory companion to SA-CVA once you're computing xVA, so we included it as part of delivering the xVA ask fully rather than narrowly

TIME: ~1.5 min

8 DATA

Data Sourcing

Section divider -- data.

- Frame this section as answering 'can we trust the inputs', which has to come before 'can we trust the model' -- a perfect model on bad data is still wrong

TIME: ~20 sec

9 DATA

Real Market Data Everywhere Verifiable; Proxies Documented Where Not

Walk the data table left to right: live feeds for the backbone, licensed snapshots for the long end and JPY, public series for validation and backtesting.

- USD curve bootstrapped from 33 live CME SOFR futures (Databento), cross-checked against Treasury.gov
- Licensed Bloomberg data used only for derived numbers (curve splice, swaption-implied mean reversion) -- never redistributed raw
- JPY OIS history (35 tenors, since 2011) provided by the project team is the most complete JPY source and drives the JPY factor

TIME: ~2 min

10 DATA

Two Real Bugs Found by Cross-Checking Against Independent Sources

This slide builds credibility: we show the actual process of catching errors, not just the final clean numbers.

- The flat long-end extrapolation understated BF_0003 (the \$500M 2049 bond forward) by about 16% -- found by repricing on Bloomberg's own curve
- USDJPY drift sign error found by deriving drifts from the martingale condition, not by chance -- the earlier martingale test only covered 6 high-vol names
- 13 total defects logged and fixed in Section 12 of the report; every one was caught by an independent cross-check, not assumed away

TIME: ~1.5 min

11 DATA**The USD Curve: 33 Live Futures Bootstrapped, Spliced at the Long End**

Show the actual curve-construction process -- this curve discounts every trade and drives the Hull-White fit.

- Each 3-month SOFR future gives an implied forward rate for its reference quarter; chaining consecutive quarters multiplies discount factors
- Futures end at about 6.5 years; beyond that the curve follows the forward structure of Bloomberg's own zero curve, joined continuously -- not flat extrapolation
- The splice matters in dollars: BF_0003's NPV moves from \$101.4M (flat) to \$121.3M (spliced), within 0.4% of repricing on Bloomberg's own curve directly

TIME: ~1.5 min

12 DATA**Volatility: 3-Year Realised, Fitted to Where the Book's Risk Actually Lives**

Volatility is the key diffusion input for every risk factor; show both the equity cross-section and the rate-vol fitting decision.

- 37 equities span an order of magnitude in realised vol (a handful of names at 50-76%) -- these dominate tail exposure wherever they appear
- USD Hull-White sigma is fitted to realised 2y-30y Treasury yield vol (96bp), not the overnight SOFR fixing (63bp) -- the overnight rate moves in discrete Fed-date steps and says little about the long end the book actually depends on
- This fix (Section 3.2) was itself one of the defects corrected: the original overnight-SOFR calibration understated 10-day 20y-yield moves (11bp modelled vs 15bp realised)

TIME: ~1.5 min

13 DATA**Correlation: One Static 39x39 Matrix From 616 Aligned Trading Days**

Explain how the correlation matrix was actually built, including the cross-market calendar-join subtlety.

- US and Tokyo trade on different calendars, so series are joined by calendar date, not timestamp -- an earlier version joined by timestamp and silently emptied ~70% of the table (defect #4)
- 741 pairwise correlations center on 0.14 -- a broad positive market block, not a few isolated pairs
- The USD rate row uses 10-year Treasury yield changes, not the overnight SOFR fixing, because the fixing moves in Fed-date steps and correlates with nothing

TIME: ~1.5 min

14 DATA**Every Assumption Not Sourced From Data, Stated Explicitly**

A single consolidated slide of every number in the engine that is an assumption, not a measurement -- full transparency in one place.

- None of these are hidden inside a formula -- each is named here and its sensitivity is shown somewhere else in this deck
- The two biggest-impact assumptions are the counterparty credit rating (drives every CVA/SA-CVA number) and whether the book is margined at all (drives the choice of exposure convention)
- Every other input in the engine -- curves, vols, correlations, trade terms -- is measured from real data, not assumed

TIME: ~1.5 min

15 PIPELINE

Pipeline and Architecture

Section divider -- pipeline.

- This section exists to show the engine is one coherent system, not a pile of scripts -- worth saying explicitly since the next two slides move fast through a lot of structure

TIME: ~20 sec

16 PIPELINE

Six Stages From Real Market Data to Every Risk Output

Walk the pipeline left to right; emphasize that the pricing library is never modified -- a deliberate design choice so errors cannot hide inside re-implemented pricing.

- Every pricer is a pure function $npv = \text{pricer.npv}(\text{MarketState})$ -- the simulation never contains its own pricing logic
- Dates use pillar tenors (O/N..10Y) plus every trade's own event date forced onto the grid -- 42 nodes for this book, denser than a plain monthly grid (18) and exact on cash-flow dates
- This is the same pipeline that produces exposure, Greeks, stress, CVA/DVA/FVA/KVA, SA-CVA and SA-CCR -- one consistent engine, not bolted-on add-ons

TIME: ~2 min

17 PIPELINE

Simulation Dates: Pillar Tenors Plus Every Trade's Own Event Date

Explain the date-grid design choice: not a plain monthly grid, but market pillar dates with every trade's own cash-flow date forced in.

- Monthly steps are simple but blunt -- they spend nodes evenly when risk actually changes fastest near-term, and a trade's own reset or maturity rarely falls on a month-end
- Following Capitolis' guidance, we adopt pillar dates: the same standard market curve tenors used to build the SOFR curve (O/N, T/N, 1W, 2W, 1M, 2M, 3M, 6M, 9M, 1Y, 18M, 2Y...) -- dense short-term, sparse long-term, mirroring production CCR practice
- Every trade's own reset/settlement/maturity date is forced onto the grid in addition, so exposure jumps at real cash-flow dates are captured exactly, not smoothed over

TIME: ~1.5 min

18 PIPELINE

Building the Engine, Steps 1-4: From Raw Data to Calibrated Models

Walk the actual build order: repo and data layer first, then the risk-factor models, then calibration, then the simulation engine that ties them together.

- Each step maps to a real Python package in the repository -- this isn't a conceptual diagram, it's the literal module structure
- Steps 1-2 are almost entirely about getting real market data right; steps 3-4 turn that data into calibrated, simulatable models
- The next slide covers steps 5-8: pricing integration through to validation and testing

TIME: ~2 min

19 PIPELINE

Building the Engine, Steps 5-8: Pricing Through to Validation

Second half of the build: integrate the given pricing library, compute Greeks, layer on credit risk and capital, then validate everything independently.

- Step 6 (pricing) deliberately reuses the Capitolis-supplied library unmodified -- a MarketState is built from simulated factors and handed to pricer.npv()
- Steps 7-8 (Greeks, exposure/credit) are built on the same simulated paths as step 5 -- not separate re-simulations, so every output is mutually consistent
- Step 9 (validation) closes the loop with checks that are independent of the engine's own outputs -- martingale tests, a parametric VaR benchmark, and a historical backtest

TIME: ~2 min

20 MODELS

Modelling Choices

Section divider -- models.

- Set expectations: this section is about the models we actually shipped, with their fitted parameters -- the trade-off section right after is where we justify them against the alternatives, so don't pre-empt that debate here

TIME: ~20 sec

21 MODELS

USD Short Rate: Hull-White 1F, Fitted Exactly to Today's Curve

This is the backbone of the whole engine: every discount factor at every scenario and node comes from this model.

- Analytic bond prices mean one function evaluation per (scenario,node) -- 10x faster than building a curve object, and exact
- Gaussian, so it supports negative rates -- essential for JPY, harmless for USD
- sigma fitted to realised 2y-30y Treasury yield vol (96bp), not the overnight SOFR fixing (63bp), because the book's risk comes from the long end

TIME: ~1.5 min

22 MODELS

Equities, FX and Correlation: GBM Driven by the Simulated Rates

Equity and FX dynamics are standard GBM, but the drifts are derived, not assumed -- and verified by martingale tests.

- Drifts follow from the martingale condition on every USD-denominated tradable asset -- including the quanto correction for JPY-listed names
- One static 39x39 correlation matrix from 616 aligned daily dates, applied via Cholesky -- the USD rate row uses 10-year-yield changes, not the overnight SOFR fixing, which correlates with nothing
- This drift derivation is exactly where the USDJPY sign error (defect #11) was caught and fixed

TIME: ~1.5 min

23 MODELS

JPY: a Genuine Negative-Rate-Capable Hull-White Factor

JPY used to just be a constant differential off the USD rate -- we built a real second factor from the actual JPY curve history.

- Real TONA history shows JPY rates were at or below zero for most of 25 years -- a model with a floor could not represent this
- Mean reversion calibrated at its lower bound (0.001) because every real JPY calibration route gives a negative value -- a genuine structural finding, not bad data
- Only 2 of 16 trades hold JPY names, so the effect is confined to their netting sets -- quantified in the attribution on the Results section

TIME: ~1.5 min

24 MODELS

Exposure Built to the Brief's Exact Definition, Not a Simplification

This slide shows we implemented the brief's definition precisely, including the subtleties (settlement-window exclusion, netting-before-max).

- $\text{exposure}(t) = \max(V(t+10\text{bd}) - V(t-1\text{bd}), 0)$, netted by counterparty before taking the positive part
- A trade settling inside a close-out window is excluded from both legs of that window -- its value drop to zero is a cash settlement, not a market move
- This was itself a defect fix (#9): the earlier version reported the uncollateralized level exposure as the headline, found by reviewing the work against the kickoff deck

TIME: ~1.5 min

25 TRADE-OFFS

Trade-Offs, Alternatives and Enhancements

Section divider -- trade-offs.

- This is the section that answers 'why not X instead' before anyone has to ask -- every alternative shown here was actually built and measured, not dismissed on intuition, which is the point to make explicit before diving in

TIME: ~20 sec

26 TRADE-OFFS

Latin Hypercube Beats Four Other Sampling Methods on a Controlled Test

This slide shows we tested five real alternatives, not just picked one by convention.

- Controlled European-call test: LHS has 50.6x lower RMSE than pseudo-random, the clear winner
- On the real 663-dimensional engine the advantages compress -- no single method wins both PFE99 and median PFE
- LHS is the only method that is both best on the controlled test AND better than pseudo-random on both real-engine statistics -- that consistency is why it was selected over Sobol, the controlled-test runner-up

TIME: ~2 min

27 TRADE-OFFS

Scenario Count: N=5,000 for Reporting, N=10,000 for Limit Sign-Off

Error falls as $1/\sqrt{N}$; we picked the count where further scenarios stop being worth the runtime.

- Monte Carlo error follows $1/\sqrt{N}$ exactly -- shown on the chart
- N=5,000 gives 0.29% relative SE at 8 minutes; N=10,000 gives 0.21% at 15 minutes for sign-off
- Beyond ~15-20k scenarios, returns diminish sharply while cost keeps scaling linearly -- and the remaining sampling error is an order of magnitude smaller than the model sensitivities in Section 7.9

TIME: ~1.5 min

28 TRADE-OFFS

Two-Factor Rates (G2++) Changes the Portfolio MPE99 by Only -3%

We implemented and fully calibrated the two-factor alternative, not just discussed it -- and it does not justify its added complexity here.

- G2++ lets the curve change slope independently of its level, but fits the realised yield covariance with a 16% Frobenius error -- it cannot reproduce the hump near 5 years
- The effect on the headline number is small: -3% on portfolio MPE99, -1% to -3% by counterparty
- One-factor Hull-White retained as the default: materially simpler, calibrates cleanly to the swaption cube, and the model-risk study (Section 7.9) shows equity volatility and rate-equity dependence matter far more than this choice

TIME: ~1.5 min

29 TRADE-OFFS

Full-Rank Correlation Kept: PCA Factor Model Adds No Speed, No Consistent Accuracy

A second genuine alternative we built and measured, and rejected as the default on evidence, not intuition. Also show what the factors themselves mean economically -- that's what 'explainability' requires.

- A 5-factor PCA model explains only 47% of variance and understates netting-set vol by up to 3.9% -- it approximates, does not replicate, the empirical matrix
- Counter to the usual argument for factor models: correlated-shock timing shows PCA is NOT faster (2.08s for 5 factors vs 1.91s full) because it trades a 39x39 matrix product for more random draws
- Factor 1 (20% of variance) is a broad market mode; factor 2 (8.9%) cleanly separates Japan from US listings -- the factors are economically readable, not just statistical artifacts
- Kept as a documented robustness option (`corr_mode='factor'`), not the default

TIME: ~2 min

30 TRADE-OFFS

Common Random Numbers Make Bump-and-Reprice 282x Less Noisy Than Independent Draws

Four Greeks methods were considered; CRN bump-and-reprice is the production method because it works on the black-box pricers for every measure including quantiles.

- Common random numbers cut the EE delta estimator's standard deviation from \$245k to \$1k -- 282x -- this, not the sampling scheme, is what controls Greeks precision
- Pathwise is validated and used as a free cross-check for equity/FX (0.2-0.4s vs ~260s) but cannot reach rates without differentiating the pricers
- Adjoint differentiation is the asymptotically fastest method but needs a differentiable port of the supplied black-box pricers -- listed as future work, not adopted

TIME: ~1.5 min

31 FEEDBACK

Feedback Incorporated

Section divider -- feedback.

- Frame this as closing the loop, not introducing new work -- every item here should feel like a callback to something the audience themselves raised, which is why several are tagged 'Feedback-driven' inline earlier in the deck too

TIME: ~20 sec

32 FEEDBACK

11 Feedback Items Across Two Rounds, All Addressed

Give the full tracker as a single visual before drilling into highlights -- this is the 'you said / we did' overview.

- Don't read all 8 rows -- let the table speak, and instead name the pattern: every item has a number attached to it, not just a description of what we did, because a client wants evidence an item is actually closed, not just acknowledged
- If pressed on any one item, you can go deeper live since every row traces to a specific section of the report -- treat this as an index you navigate from, not a wall of text to narrate
- Good moment to pause and ask if anyone wants to go deeper on a specific item before moving to items 9-11 on the next slide

TIME: ~1.5 min

33 FEEDBACK

Items 9-11: CVA Sensitivities, DV01 via a Jacobian, Excel Parametric Check

Close the tracker with the three most technical items -- the DV01 Jacobian fix is the single biggest methodology correction from feedback.

- DV01 Jacobian fix moved 28 points of portfolio DV01 attribution from the 10y bucket into the 30y bucket where the 2049 bond forward actually lives
- The Excel parametric check (item 11) is an independent, non-Monte-Carlo validation that the audience can reproduce themselves
- Every item on this tracker is also tagged 'Feedback-driven' inline where it reappears in the Models, Trade-offs and Results sections

TIME: ~1.5 min

34 FEEDBACK

DV01 via a Jacobian: Full Before/After, and Why It Changes Hedging

Go deep on item 10 -- this is the single biggest methodology correction from feedback, and it has a real hedging consequence, not just a cosmetic one.

- A 'Jacobian' here means the DV01 to each of the curve's own ~45 construction instruments (SOFR futures + Bloomberg long-end points), not to 8 interpolated zero-rate tenors
- The old method's 10y bucket (39% of DV01) was an artifact of where the zero-rate grid happened to sample the fitted curve, not of where the book's rate risk actually sits
- Practical consequence: a hedger following the old attribution would buy/sell 10y instruments to offset CPTY_C's rate risk - but 82% of the real risk is in the 30y bucket, where the 2049 bond forward actually lives, leaving the position effectively unhedged against a curve twist

TIME: ~2 min

35 RESULTS

Results

Section divider -- results, the core of the deck.

- Signal the pace change here -- everything before this was setup; this section is where the audience's actual questions ('what's the number, why, how sure are we') get answered, so expect it to run longer and take more questions along the way

TIME: ~20 sec

36 RESULTS

Peak Portfolio MPE99 is \$51.0M, 78% of the Sum of Counterparty MPE99s

Restate the headline with full precision and immediately explain the diversification number -- this is the single most important slide in the deck.

- The 78% diversification figure is the one number on this slide worth explaining rather than reading: it's low specifically because netting is only ever within a counterparty, never across -- so three separate books peaking in the same three-week window barely offset each other at all. That's a structural fact about the book, not a weakness in the model
- Anticipate the obvious follow-up: 'is \$51.0M a big number?' -- frame it relative to today's \$129.8M uncollateralized exposure, since that comparison is what tells the client how much the close-out/margin convention is doing for them
- This is the slide to slow down on -- everything later in Results (attribution, stress, Greeks, credit) is explaining different facets of this one number, so make sure the room has it before moving on

TIME: ~2 min

37 RESULTS

Exposure Runs Off Within Four Months as the Book's Trades Mature

Show the time profile -- almost all counterparty credit risk sits in the next four months.

- Nearly the whole book matures within ~4 months; only BTRS_0001 and EQTRS_0008 run past mid-2027
- Portfolio EE falls by more than 90% by end of December 2026
- This is why monitoring should concentrate on the next four months, not a flat one-year horizon

TIME: ~2 min

38 RESULTS**Per-Trade Detail: BF_0003 Alone Drives CPTY_C's \$29.6M Peak MPE99**

Drill from counterparty to trade level -- shows the per-trade numbers do not simply sum to the counterparty figures.

- Worth flagging explicitly: if someone adds up the per-trade MPE column they will NOT get the counterparty total from the earlier slide, and that's correct, not an error -- the 99th percentile of a sum is never the sum of 99th percentiles, because the trades don't all hit their own worst case on the same date
- This table is really here to answer one question in detail: which individual trades are actually driving each counterparty's number, since the netting-set view alone can't tell you that
- If the audience starts asking about a specific trade not shown (there are 16, only 8 fit), offer to pull up the full table from the report rather than reciting numbers from memory

TIME: ~1.5 min

39 RESULTS**Why the Portfolio is Close to A+C, Not A+B: All Three Are Pay-Equity**

Answers feedback item 4 directly with the evidence -- A is equity risk, C is rate risk plus meaningful equity, and all three share a common driver.

- Every netting set holds pay-equity swaps, so all three gain when the equity market falls -- that common exposure is why dependence is positive
- CPTY_C is not a pure rate netting set: DV01 \$594k/bp dominated by BF_0003, but its equity delta (\$1.24M) is comparable to CPTY_B's
- At the portfolio's peak date, standalone PFE99 are $A \$25.9M + C \$28.5M = \$54.4M$ (107% of the \$51.0M portfolio figure) - the portfolio is close to A+C, with B adding little

TIME: ~2 min

40 RESULTS**Corrections Moved the Headline From \$45.7M to \$51.5M, One Fix at a Time**

This is the attribution waterfall -- every number change is traced to a specific, named correction, run on identical random numbers.

- Each correction introduced one at a time on the same 2,000 scenarios and random draws, so differences are attributable, not noise
- Largest single move: the long-end volatility fit and 10-year-yield correlation correction (+\$4.9M portfolio), because it re-derives sigma from where the book's risk actually sits
- Monte Carlo noise between runs of different size is a few percent, so only differences larger than that should be read as a real correction effect

TIME: ~1.5 min

41 RESULTS

Model Risk: Equity Volatility and Rate-Equity Dependence Dominate, Not the Rate Model

This ranks the real sources of model uncertainty -- important for the audience because it tells them where to focus review effort.

- A 25% rise in equity/FX volatility moves portfolio MPE99 by +19%; a 30%-of-the-way-to-one move in equity correlations moves it by +22%
- The two-factor rates model (G2++) changes the headline by only -3% -- so rate-model choice is a minor source of uncertainty here
- Each row is a one-at-a-time perturbation on 2,000 scenarios, same random numbers -- not a vague sensitivity statement

TIME: ~1.5 min

42 RESULTS

Book Greeks Today: DV01 of \$622k/bp Dominated by the CPTY_C Bond Forward

The t=0 Greeks by netting set, validated against analytic values -- sets up the DV01 Jacobian fix on the next data point.

- Make the sign explicit for the room: a negative equity delta means we gain when equities fall, which is the mirror image of being pay-equity on every swap -- worth stating in plain English since the table alone just shows a negative number
- These are t=0 snapshot Greeks, not scenario-based risk measures like the MPE99 shown earlier -- they tell you today's instantaneous sensitivity, which is a different (and complementary) question from 'how bad could it get'
- This table is the setup for the next topic (DV01 via a Jacobian) -- worth a one-line bridge: 'that \$594k DV01 number for CPTY_C is correct in total, but where it sits by tenor is what feedback item 10 corrected'

TIME: ~1.5 min

43 RESULTS

Validated to 0.0000%, Then Stress-Tested Across 13 Scenarios

Two forms of checking the engine in one slide: independent validation of correctness, then a structured stress programme (feedback item 1) showing how the results move under shocks.

- Validation: t=0 self-check to 0.0000%, martingale tests, a 1.04x delta-normal VaR benchmark, and 142 automated tests -- each isolates one layer of the engine, independent of the headline results
- Stress: 13 scenarios (8 hypothetical + 5 historical, incl. 2008 GFC) run as a full Monte Carlo re-run from the shocked state, same random numbers as the base case
- Close-out MPE99 moves at most -24%/+23%; level MPE99 moves up to +86% -- margin is why the close-out measure is far less sensitive to instantaneous shocks

TIME: ~2.5 min

44 RESULTS

Backtest: 99% Quantiles Close to Calibrated, 4/6/2 Exceptions vs 2.4 Expected

Kupiec backtest against realised history, out of sample -- CPTY_B is the one borderline case, consistent with GBM understating fat tails.

- Equity leg: exceptions 4/6/2 (A/B/C) against 2.4 expected over 242 non-overlapping 10-day windows at the 99% level
- Kupiec test does not reject for 2 of 3 netting sets at the 5% significance level; CPTY_B (6 exceptions) is borderline -- consistent with GBM understating fat tails
- Rates leg: the long-end yield-vol fit (96bp) is the more stable calibration across the 2020-2026 backtest window than the overnight-SOFR calibration (63bp)

TIME: ~1.5 min

45 RESULTS

CVA \$11.7k / DVA \$11.2k / FVA \$0.5k / KVA \$22.4k on the Close-Out Convention

The full xVA suite, on both exposure conventions, all consistent because they reuse the same simulated paths.

- Close-out: CVA and DVA nearly cancel (\$11.7k vs \$11.2k) because both are 10-day moves around a margined start; KVA (\$22.4k at 10% CoC) exceeds CVA because it scales the whole EPE profile, not a small IG default probability
- Uncollateralized level exposure is far larger: CVA \$247k, dominated by CPTY_C's \$121M bond forward sitting at risk with no margin
- CVA proxy is BBB; CVA scales roughly linearly with spread -- AAA \$5.0k to B \$32.8k, a 2.6x range between AA and BB

TIME: ~1.5 min

46 RESULTS

Extra Credit: Pricing a Risky Bond With an Issuer-Credit Proxy

The second kickoff extra-credit item -- a sample risky-bond trade outside the 16-trade book, pricing issuer credit risk directly into the bond.

- Issuer credit and counterparty CVA are two different things: issuer credit lowers the bond's own value; counterparty CVA (shown on the previous slide) is the risk that the other side of the trade defaults
- No CDS quotes were obtainable for the issuer, so the same proxy method as the counterparty spreads is reused: ICE BofA BBB option-adjusted spreads with the pricing library's credit-triangle survival curve
- The risky bond is worth \$934k less than the risk-free version -- issuer credit is not a small effect here, it flips the forward's sign (from +\$654k to -\$280k)

TIME: ~1.5 min

47 RESULTS

SA-CVA Capital \$0.10M (Close-Out) / \$2.23M (Level); SA-CCR EAD \$260.3M

Regulatory capital view -- both computed, both dominated by counterparty credit spread risk, not by market risk.

- SA-CVA requirement is dominated by counterparty credit spread delta (88% of the total) -- a 5% risk weight on spreads is large relative to a ~100bp spread level
- SA-CCR EAD \$260.3M is a different quantity from Monte Carlo MPE99: replacement cost is today's full MTM, add-on is a one-year supervisory approximation -- included as an independent order-of-magnitude check and the regulatory view of the same book
- RWA (12.5x capital): \$1.2M close-out, \$27.8M uncollateralized

TIME: ~1.5 min

48 RESULTS

CVA Sensitivity Numbers: Credit Spread Delta Is 18x the Rate Delta

This is the raw sensitivity data behind the previous slide's chart and capital table -- the actual CVA01/CS01-style numbers, not just the aggregated capital.

- These are the Basel bucket-level deltas (pre risk-weight) that feed directly into the SA-CVA capital calc on the previous slide -- apply the 5% credit-spread risk weight and bucket correlation and \$1.99M becomes the \$85.2k close-out capital shown there
- Credit spread delta dominates rate delta by about 18x, and within credit spread delta, 95% sits in the 0.5y and 1y buckets because CPTY_C's \$121M bond forward matures inside a year
- This directly answers feedback item 9: CVA is for understanding sensitivities, not just a single number -- these are the per-tenor, per-counterparty building blocks

TIME: ~1.5 min

49 CONCLUSIONS

Conclusions and Next Steps

Section divider -- conclusions.

- Validated, reproducible engine; limitations disclosed; next steps depend on data Capitolis holds, not more engineering

TIME: ~20 sec

50 CONCLUSIONS

Validated and Reproducible; Limitations Disclosed, Not Hidden

Close with both sides honestly -- what we're confident in, and what still depends on data or assumptions.

- Every limitation here has a quantified impact shown earlier in the deck -- none is a vague caveat
- The single biggest open question is whether any trade is margined -- MPOR illustration shows it would move CPTY_C's MPE99 by 82%
- This is deliberately a balanced slide: confidence in engineering, honesty about what the data cannot tell us

TIME: ~2 min

51 CONCLUSIONS

Recommended Next Steps: Confirm Margin Terms First

End on the single highest-leverage open question, then the concrete roadmap items in priority order.

- Confirming whether any trade is margined matters more than any further modelling refinement -- it changes CPTY_C's number several-fold
- Real counterparty ratings or CDS would remove the single largest assumption in the credit numbers
- Everything else listed is scope and data, not engineering -- the plan as given has been fully executed

TIME: ~1.5 min

52 CONCLUSIONS

Thank You

Close and open the floor -- the headline number and the one open question (margin terms) are worth repeating before questions start.

- Headline: portfolio MPE99 \$51.0M on the brief's close-out definition, validated to 0.0000% self-consistency, 142 automated tests passing
- The single highest-priority follow-up is confirming margin terms -- it moves CPTY_C's MPE99 by up to 82%
- Appendix slides are available if anyone wants to go deeper on a specific topic -- decision register, data lineage, G2++/PCA detail, or the reproduction commands

TIME: ~open for questions

53 APPENDIX

Appendix

Transition slide only -- skip past this unless a question calls for one of the backup slides.

- 10 appendix slides: full decision register, data lineage, 2008/2015 stress detail, per-trade stress heatmap, DV01 bucket detail, G2++ calibration, PCA loadings, backtest rates detail, code map, reproduction commands
- Not walked through live -- jump to the relevant one only if a question needs it

TIME: ~skip unless asked

54 APPENDIX

A1. The Full Decision Register (Report Section 10)

One-slide index of every modelling choice in the deck and what else was considered -- useful if asked 'did you consider X'.

- Every row maps to a trade-off slide already presented in Section 6 -- this is the compact summary, not new material
- Use this to answer 'why didn't you use X' questions by pointing at the specific row

TIME: ~if asked

55 APPENDIX

A2. Full Data Lineage (Report Appendix B)

Full source-by-source data inventory -- use if asked exactly where a specific number came from.

- Every input is either live/public (reproducible by anyone) or licensed Bloomberg/project-team data (derived numbers only, never redistributed raw)
- This is the detailed version of the summary already shown in Section 3

TIME: ~if asked

56 APPENDIX

A3. 2008 GFC and 2015 China Devaluation: Full Detail

Backup detail on the 2008/2015 historical scenarios added for feedback item 6.

- 28 of 37 names have price history back to 2007
- 2008: -23% median / -44% worst single name; moves close-out MPE99 -13%, level +56%

TIME: ~if asked

57 APPENDIX

A4. Per-Trade Stress Sensitivity Heat Map (Report Figure 34)

Visual drill-down of the portfolio stress table (Section 8.1) to the individual-trade level -- which trades move the most under which scenario.

- Darker cells are larger percentage moves in that trade's exposure under that scenario -- BF_0003 and the equity TRS names are the hottest rows, consistent with the concentration story told throughout the deck
- Use this if asked which specific trade drives a given scenario's portfolio-level number

TIME: ~if asked

58 APPENDIX

A5. DV01 Bucket Detail: Old Zero-Grid vs New Par-Instrument Jacobian

Full numeric backup for the DV01 Jacobian slide already presented in Section 7 -- same table, in case someone wants to check the arithmetic live.

- Both methods' bucket sums match their own parallel-bump check to within 0.05% -- the correction was about attribution across tenor, not total risk
- Point to the 10y/30y rows specifically if asked to justify the hedging consequence described in Section 7

TIME: ~if asked

59 APPENDIX

A6. G2++ Two-Factor Rate Model: Full Calibration

Full calibration detail for the G2++ alternative referenced in Section 6 -- have this ready if asked for the actual fitted parameters.

- 16% Frobenius fit error is the headline number to repeat -- it's why G2++ isn't the default despite matching level and slope
- $\rho = -0.95$ (strongly negative) is what lets two positively-mean-reverting factors combine into a curve with both a level and a slope component

TIME: ~if asked

60 APPENDIX

A7. PCA Factor Loadings: What the Factors Actually Are

The economic reading behind the PCA trade-off slide in Section 6 -- have this ready if asked what the statistical factors actually represent.

- Factor 1 (20% of variance) is a broad market factor across both US and Japan names -- the usual first PCA factor in any equity book
- Factor 2 is cleanly a Japan-vs-US split, which is reassuring: a statistical factor lining up with an economic grouping is a sanity check that the PCA is picking up real structure, not noise

TIME: ~if asked

61 APPENDIX

A8. Backtest: Rates Leg Detail (10-Day Yield-Change Quantiles)

Backup for the backtest slide in Section 8 -- the rates-only quantile comparison behind the two sigma calibrations discussed there.

- This is why the long-end fit sigma (0.96%) was chosen over the raw overnight-SOFR sigma for the headline run -- its exception rates are closer to nominal across all three tenors
- Use this if asked to justify the sigma calibration choice with the actual exception counts rather than just the summary statistic

TIME: ~if asked

62 APPENDIX

A9. Code Map (Report Appendix D)

Codebase orientation for a technical reviewer -- use if asked how the engine is organized or where a specific piece of logic lives.

- Modular by design: each package has one job (market data, models, simulation, Greeks, stress/validation, exposure) and is independently testable
- 142 tests target these modules directly, not just end-to-end -- that's what makes the 'validated' claim in the conclusions credible

TIME: ~if asked

A10. Reproducing Every Result (Report Appendix E)

Closing technical-credibility slide -- every number in this deck comes from one of these nine commands, nothing is hand-tuned or one-off.

- Each script maps to one deliverable: simulation profiles, Greeks, CVA/SA-CVA, xVA/SA-CCR, stress, backtest, and the full test suite
- Offer this as the answer to 'how do we know these numbers are real' -- anyone with the repo and the data can regenerate every figure in the report

TIME: ~if asked